OPEN SOURCE · CANADA (US EXPERIMENTAL) · RUNS ON YOUR MACHINE

taxjson

Your broker's CSV in. Schedule 3 out. Capital gains, dividends and the superficial-loss rule computed the way the CRA defines them — across every account you own — with every rule citing its source.

```
bash -c "$(curl -fsSL https://taxjson.com/install.sh)"
```

Installs `taxjson` and its short name `tjs` · taxjson.com · github.com/taxjson/taxjson · MIT

THE PROBLEM

The slip your broker files is not your cost basis

A T5008 reports proceeds per account. The Income Tax Act taxes you on **one average cost per security across all your accounts**, denies losses when any affiliated account — your RRSP included — bought the same thing within 30 days, and folds option premiums, splits, spin-offs and return of capital into that cost. Nobody's statement does that math for you.

Cross-account pooling

s.47: one ACB pool per identical property. A US listing and its TSX listing are the same pool; a CDR is not.

Superficial losses

s.54 / s.40(2)(g): a loss is denied if you *or an affiliated person* bought within ±30 days and still holds at day 30 — sized sale by sale. Denied into an RRSP, it is gone for good.

Everything that moves cost

Assignments (IT-479R), warrant exercises, splits, renames, spin-offs (s. 86.1), mergers with elections, ROC, reinvested distributions — in the right order, on the settlement date.

Spreadsheets get the easy 90% right. The other 10% is where reassessments come from.

WHAT IT DOES

Three steps, all on disk

inputs/

Every year's broker CSV exports, one folder per account: IBKR, Questrade, RBC, Webull, Kraken, Coinbase, any other broker through the generic importer — plus plain-text `.tt` lines (an `OPENING` balance, an older purchase)

→

tjs run

Parse · convert at the Bank of Canada rate per settlement date · pool identical securities · dated renames and corporate actions with elections · superficial-loss pass across taxable and registered accounts

→

reports/

Per-account summaries, Schedule 3 / Form 8949 rows, the wash radar, T1135, a tax and minimum-tax estimate — and the broker cross-check

Year folders

One `inputs/` every year shares, and a folder per year: `taxjson.toml` (year, country, accounts, the figures you enter by hand), `ticker.map` (your symbol rules), `holdings/`. In Canada one `tobase.map` beside them: the interlisted pairs every year reads. `tjs new-year` starts the next.

Row accounting

Every parser accounts for every row it reads. Rows it recognises but does not translate are counted and named — never silently dropped.

Traceable numbers

`tjs audit` recomputes every disposition from the parsed broker row through FX, ACB and the wash determination, and ties it out against the reports.

RULES WITH RECEIPTS

Every computation names its source

RULE	WHAT TAXJSON DOES	SOURCE
Identical properties	One average-cost pool per security across accounts; cross-listings merged, CDRs kept apart; a ticker change carries the pool on its date	s.47(1); IT-387R2; T4037
Superficial loss	CRA's formula, sale by sale: the least of the units sold, bought in the 61-day window, and held at day 30; the RRSP/TFSA is affiliated; denied-into-plan is permanent	s.54, s.40(2)(g)(i), s.53(1)(f), s.251.1(1)(g)
Timing	Sales on the settlement date, including at year end; a dividend in the year paid, a Canadian trust's distribution in the year payable	T4037; s.82(1); s.104(13)
Foreign currency	Bank of Canada rate on each settlement date, never an annual average	Folio S5-F4-C1
Options	A written option's premium is a gain in the year written; a buy-back is a loss in its own year; an assignment folds the premium into the shares; an exercised warrant's cost goes into the shares too. `option-boundary` says when a filed year needs a T1-ADJ	s.49(1)–(4); IT-479R
Corporate actions	Splits, spin-offs, mergers with the rollover election recorded and a filing reminder at every run; ROC reduces ACB	s.86.1, s.85.1, s.53(2)(h)
US (experimental)	FIFO; §1091 wash sales with basis and holding-period carry, IRA repurchases permanent; lots carried between your own accounts; §355 / §356 per lot; Form 8949 codes	§1091, Rev. Rul. 2008-5; Reg. 1.358-2, 1.356-1(b)

`tjs tax-logic` prints every rule taxjson applies, one line each, with the id its tests cite — Canada's and the USA's apart, never mixed. REFERENCES.md maps each to its source, including what is deliberately *not* modelled. A rule that disagrees with its source is a bug; that is the most useful issue the project can receive.

WORKED EXAMPLE

A loss you cannot claim

What happened

The demo project's `ib` account sells 10 NVDA at a loss on November 20, 2024, buys 10 back on December 5 and still holds them on December 20.

What the slip says

A capital loss of 64.71. The spreadsheet agrees.

What the Act says

The least of 10 sold, 10 bought in the 61-day window and 10 held at its end: all superficial (s.54); the loss moves into the new shares' cost. Bought back in the RRSP, as INTC is, it is denied for good.

```
$ tjs wash-sales
SUPERFICIAL LOSSES — CAD, tax year 2024, basis: wash-adjusted
DENIED: the loss the superficial-loss rule (s.54) denies; ALLOWED: the loss you claim.
ACCOUNT  DATE        SYMBOL   QTY  PROCEEDS      COST     GAIN  DENIED  ALLOWED
-------------------------------------------------------------------------------
margin   2024-08-05  INTC.US   20    564.17    966.96  -402.78  402.78     0.00
ib       2024-11-20  NVDA.US   10  1,953.00  2,017.71   -64.71   64.71     0.00

DENIED:    467.50 CAD over 2 superficial loss(es) (402.78 permanently denied)
DEFERRED:  64.71 CAD in open positions' cost — tjs list
```

Only a tool that sees every account at once can get this right. That is the whole design.

VERIFICATION

Checked against the one thing that cannot lie

Internal reports can all agree and still be wrong. `tjs sanity` compares the positions the books say you hold with the positions your broker's own export says you hold, account by account — every run ends with it once the snapshots are in `holdings/`, matched to their accounts without a setting. Any difference is a trade that has not reached the CSV yet, or a booking problem — there is no third kind.

12,500+ tests

Every fix lands with a regression test that failed before it, tagged with the tax-logic rule it pins; the gate fails on a rule with no test.

Fuzzers and mutants

Three property fuzzers run thousands of random ledgers per run; mutation testing of both engines finds the boundaries no test pins.

Two full audits

Eight review rounds, a security review and two full-coverage audits of the whole codebase. Every confirmed finding reproduced, fixed, pinned.

Published limits

docs/limits.md lists what is out of scope and what to do by hand, with the citation.

COMMANDS

One pipeline, a few dozen views on it

Set up

`checklist` `init` `years` `new-year` `align` `format` `format-map` `migrate` `fetch` `elect` `ticker-map` `update-tobase-map`

Build the books

`run` `compare` `snapshot` `crypto-sends` `find-missing-history` `opening`

Summaries

`amt` `div-estimate` `estimate` `fx-cash` `instalments` `option-yield` `plot-tax` `stats` `sum` `win-rates`

Positions

`list` `shares`

Row listings

`cash` `dil` `divs` `events` `fees` `gains` `leaps` `roc` `staking` `trades` `transfers`

Totals by type

`ccd-sum` `dil-sum` `divs-sum` `fees-sum` `leaps-sum` `losers` `roc-sum` `staking-sum` `trades-sum` `winners`

Before you trade

`wash-radar` `buy-check` `sell-check` `rebuy` `harvest` `tips` `watch`

Before you file

`form-export` `t1135` `reconcile-slips` `slip-audit` `carryover` `option-boundary` `close-year` `check-filed` `handoff`

Explain and check

`audit` `wash-sales` `tax-logic` `edge-cases` `check-dates` `check-tt` `sanity` `journals` `renames` `spinoffs` `splits`

Tools

`redact` `help`

The groups are the help page's (`scripts/deck_commands.py` renders them): `tjs` with no command lists every command under them, one line each — inside a project only your country's. Every command takes `-h`; the default output is the essentials, `--details` the rest; most take `--json`.

PRIVACY AND SECURITY

Nothing personal leaves the machine

Local by construction

Parsing, pooling and the wash pass run on your computer — no server, no web UI, just commands and files. A run goes online only for what its cache lacks: FX rates (Bank of Canada; Yahoo before May 2007 and for other bases), the price of an unpriced crypto row, the close for an in-kind move to or from a registered account — a pair or a symbol and a date, nothing about your books. `TAXJSON_OFFLINE=1` turns every lookup off; SECURITY.md lists every egress.

No credentials in the core

The core holds no broker client and reads no credential. Broker fetch is a separate package (taxjson-fetch, installed by default; `--without-fetch` leaves it out): it uses your own token file, never writes it into a project, and sends it nowhere but that broker.

The project guards itself

A personal-data scan runs in every gate and as the pre-push hook, fails closed, and reads a private denylist kept outside the repository; the hook also refuses a commit message that quotes a money total. `redact` keeps every row shape and drops the identifiers it recognises — review before sharing.

A standard-library core (tomli before Python 3.11); the installer adds yfinance for the rate fallback and the fetch plugin. Trustworthy before clever.

SCOPE

Canada is the product. The US engine is a draft.

Canada — supported

Average-cost ACB, the superficial-loss rule across accounts, settlement-date timing, options and warrants, dated renames, corporate actions with elections, T1135, Schedule 3, a marginal-rate estimate with provincial tables, minimum tax line by line, instalments, and carry-forwards from one year's lock to the next. Validated against two years of real multi-account books and the brokers' positions.

US — experimental

FIFO, §1091 wash sales across your accounts, lots carried between your own accounts, §355 / §356 per lot, short/long-term, Form 8949 codes and a TXF export. Implemented and unit-tested, never validated on a real account. It says so at every run. Real US exports — redacted — are the way to change that.

The two rule sets never mix: a project has one country, and a setting, flag or command the other country owns is refused, not silently applied.

Honest labelling costs nothing. A tax tool that overclaims loses the only thing it has.

BROKERS

Reads the exports you already have

BROKER	FORMAT	NOTES
Interactive Brokers	Activity Statement CSV (or a Flex query)	Trades, options, corporate actions, warrant exercises, transfers, fees, dividends; Flex auto-fetch (taxjson-fetch plugin)
Questrade	Transaction history (Excel, converted to CSV)	Trades, DRIP, cash-in-lieu, transfers; REST auto-fetch and live positions (taxjson-fetch plugin)
RBC Direct Investing	Activity export CSV	Trades, dividends with withholding, warrant exercises
Webull	Trading Summary CSV	Trades and options (BUY/SELL rows only); no income — T5 interest and dividends go in as `.tt` lines
Kraken	Trades **and** Ledgers CSV	Trades, rewards, withdrawals as custody evidence
Coinbase	Transaction history CSV	Trades, rewards, sends
Any other broker	Generic CSV mapping · `.tt` plain text	A column mapping for its CSV; ten-line files a human can write and read

Help wanted: Wealthsimple, TD Direct, BMO InvestorLine, CIBC Investor's Edge, Scotia iTRADE, National Bank Direct, Qtrade, Desjardins. A parser is an afternoon's work given one real export — `tjs redact` makes sharing one safe. An auto-fetcher for another broker is a plugin of its own.

INSTALL

Your first books in ten minutes

```
bash -c "$(curl -fsSL https://taxjson.com/install.sh)"
```

Installs the stable release into its own Python environment and links `taxjson` and its short name `tjs` onto your PATH. Needs git and Python 3.10+, nothing else. Re-run to upgrade. The Questrade / IBKR auto-fetch plugin comes with it (add `--without-fetch` to leave it out).

```
# try it first on made-up exports (Canada, 2024)
tjs init --demo ~/taxjson-demo && cd ~/taxjson-demo/2024
tjs run && tjs sum && tjs checklist --run

# your own books
mkdir -p ~/taxes && cd ~/taxes
tjs init --country canada   # or usa; inputs/ for every year, 2026/ for this one
cd 2026                     # each year's folder: its taxjson.toml, ticker.map, results
tjs checklist --run         # every step: what is done, what is next
# drop broker CSVs, all years together, into ../inputs/margin/, ../inputs/rrsp/, …
tjs run                     # books, reports, wash pass, broker check
tjs sum                     # the year, by Schedule 3 line
tjs wash-radar              # "can I sell this at a loss today?"
tjs checklist --run         # again at year end: ready to file?
```

WHERE IT GOES

Roadmap

More Canadian brokers

Driven by redacted real exports from users. One parser per afternoon, each with the row-accounting contract; auto-fetch for more brokers as plugins.

Rules retail investors hit

Non-eligible dividends and T3 box splits, the s.39(4) election, CNIL — each stated in tax-logic with its tests, whichever real exports show first.

US validation

The engine exists; it needs a real account. The label comes off when the numbers have been checked against one.

1.0

PyPI, a frozen CLI, and the interchange contract with portfolio tools written down.

taxjson.com · github.com/taxjson/taxjson · MIT · not tax advice — a calculator that shows its work.